

Linear Algebra Basics

PART A — VECTORS

1. Which of the following represents a vector?

- A. 25
- B. $[2, 5, 7]$
- C. $2 + 5 + 7$
- D. $x = 10$

2. If $x = [2, 3]$ and $y = [4, 5]$, what is $x + y$?

- A. $[6, 8]$
- B. $[8, 15]$
- C. $[2, 2]$
- D. $[4, 3]$

3. If $x = [2, 3]$, what is $4x$?

- A. $[6, 7]$
- B. $[8, 12]$
- C. $[2, 12]$
- D. $[4, 3]$

4. What does the magnitude of a vector represent?

- A. Its direction
- B. Its length
- C. Its number of elements
- D. Its number of dimensions only

5. What is the Euclidean norm of the vector $[3, 4]$?

- A. 3
- B. 4
- C. 5
- D. 7

6. Two vectors are orthogonal when:

- A. They have the same magnitude
- B. Their dot product is zero
- C. Their values are identical
- D. Their magnitudes are zero

7. What is the dot product of $[1, 2]$ and $[3, 4]$?

- A. 5
- B. 7
- C. 10
- D. 11

8. If two non-zero vectors have a dot product of zero, they are:

- A. Parallel
- B. Orthogonal

- C. Identical
- D. Linearly dependent

PART B — MATRICES

9. A matrix with 4 rows and 3 columns has dimensions:

- A. 4×3
- B. 3×4
- C. 12×1
- D. 4×4

10. Which of the following is a square matrix?

- A. 2×3
- B. 3×2
- C. 4×4
- D. 5×3

11. What does transposing a matrix do?

- A. Multiplies all elements by -1
- B. Interchanges rows and columns
- C. Converts all elements to zero
- D. Divides every element by its mean

12. If matrix A has dimensions 3×5 , what are the dimensions of A^T ?

- A. 3×5
- B. 5×3
- C. 3×3
- D. 5×5

13. Which pair of matrices can be added?

- A. 2×3 and 3×2
- B. 2×3 and 2×3
- C. 3×4 and 4×3
- D. 2×2 and 3×3

14. Matrix multiplication AB is possible when:

- A. Rows of A = rows of B
- B. Columns of A = columns of B
- C. Columns of A = rows of B
- D. Both matrices are square

15. If A is 3×4 and B is 4×2 , what is the dimension of AB?

- A. 3×2
- B. 4×4
- C. 2×3
- D. 3×4

16. If A is 2×3 and B is 3×4 , what is the dimension of AB?

- A. 2×4
- B. 3×3

- C. 4×2
- D. 2×3

17. Which statement about matrix multiplication is generally true?

- A. $AB = BA$ always
- B. AB and BA are always defined
- C. AB is not necessarily equal to BA
- D. Matrix multiplication is only possible for square matrices

PART C — LINEAR COMBINATIONS

18. A linear combination of vectors involves:

- A. Only multiplication
- B. Only addition
- C. Scaling vectors and adding them together
- D. Multiplying matrices only

19. Suppose:

$$v_1 = [1, 0]$$

$$v_2 = [0, 1]$$

Which vector can be represented as a linear combination of v_1 and v_2 ?

- A. $[2, 3]$
- B. $[1, 1]$
- C. $[5, -2]$
- D. All of the above

20. If:

$$v_2 = 3v_1$$

then v_1 and v_2 are:

- A. Orthogonal
- B. Linearly dependent
- C. Linearly independent
- D. Perpendicular

21. A set of vectors is linearly independent if:

- A. Every vector has the same magnitude
- B. No vector can be represented as a combination of the others
- C. All vectors are perpendicular
- D. All vectors contain positive values

22. If one feature can be exactly calculated from another feature, the two features are:

- A. Orthogonal
- B. Independent
- C. Redundant
- D. Random

PART D — RANK

23. The rank of a matrix represents:

- A. Total number of elements
- B. Number of rows
- C. Number of linearly independent rows or columns
- D. Number of diagonal elements

24. A matrix whose columns are all multiples of one another will have:

- A. Rank 0
- B. Rank 1
- C. Rank equal to the number of columns
- D. Infinite rank

25. If a dataset has 10 features but some features are completely redundant, the effective number of independent directions may be:

- A. Greater than 10
- B. Exactly 10
- C. Less than 10
- D. Infinite

PART E — GEOMETRIC INTERPRETATION

26. A vector can be interpreted geometrically as:

- A. A point only
- B. A direction and magnitude
- C. A scalar value only
- D. A matrix

27. What happens when a vector is multiplied by a scalar greater than 1?

- A. Its direction always changes
- B. Its magnitude increases
- C. It becomes orthogonal
- D. It becomes zero

28. Multiplying a vector by -1:

- A. Changes its magnitude to zero
- B. Reverses its direction
- C. Makes it orthogonal to every vector
- D. Removes its dimensions

29. The angle between two vectors can be determined using:

- A. Their dot product and magnitudes
- B. Their number of elements only
- C. Their matrix dimensions
- D. Their rank only

30. If two vectors point in exactly the same direction, the angle between them is:

- A. 0°
- B. 45°
- C. 90°
- D. 180°

PART F — PROJECTION

31. The projection of one vector onto another can be understood as:

- A. The component of one vector along the direction of another
- B. The difference between two vectors
- C. The magnitude of both vectors multiplied together
- D. A completely new random vector

32. If vector a is projected onto vector b, the projection tells us:

- A. How much of a lies in the direction of b
- B. How many elements are in b
- C. The rank of a
- D. The determinant of b

33. If two vectors are orthogonal, the projection of one onto the other is:

- A. Maximum
- B. Zero
- C. Equal to its magnitude
- D. Always one

34. Why is projection useful when representing data?

- A. It can represent information using a chosen direction
- B. It always increases dimensionality
- C. It removes all information
- D. It changes categorical data into numerical data

PART G — EIGENVALUES AND EIGENVECTORS

35. An eigenvector of a matrix is a vector whose:

- A. Direction remains unchanged after the transformation, apart from scaling
- B. Magnitude always becomes zero
- C. Values always become positive
- D. Number of elements changes

36. In the equation:

$$A v = \lambda v$$

what is v?

- A. Matrix
- B. Eigenvalue
- C. Eigenvector
- D. Determinant

37. In the equation:

$$A v = \lambda v$$

what is λ ?

- A. Eigenvector
- B. Eigenvalue
- C. Matrix rank
- D. Matrix inverse

38. If an eigenvalue is 5, what does this generally indicate about its associated eigenvector?

- A. The vector is rotated by exactly 5°
- B. The vector is scaled by a factor of 5
- C. The vector becomes zero
- D. The vector becomes orthogonal

39. If an eigenvalue is zero, then:

- A. The associated transformed vector is zero
- B. The eigenvector does not exist
- C. The matrix must be identity
- D. The vector becomes larger

40. Eigenvectors are useful for identifying:

- A. Important directions associated with a matrix transformation
- B. Missing values in a dataset
- C. Number of observations
- D. Categorical variables

PART H — COVARIANCE AND VARIANCE

41. Variance measures:

- A. The average value of a variable
- B. How much a variable varies around its mean
- C. The number of observations
- D. The relationship between two different variables

42. If all values in a variable are exactly the same, its variance is:

- A. 0
- B. 1
- C. -1
- D. Infinite

43. Covariance measures:

- A. How two variables vary together
- B. The mean of one variable
- C. The number of features
- D. The magnitude of a vector

44. A positive covariance generally indicates:

- A. Both variables tend to increase together
- B. One variable always decreases
- C. There is no relationship
- D. Both variables are constant

45. A negative covariance generally indicates:

- A. Both variables increase together
- B. One variable tends to increase when the other decreases
- C. Both variables are identical
- D. Both variables have zero variance

46. If two variables have covariance close to zero, it indicates:

- A. They definitely have no relationship of any kind
- B. There is little or no linear relationship detected
- C. They must be identical
- D. They must be orthogonal vectors

PART I — APPLIED LINEAR ALGEBRA

47. Consider a dataset represented as a matrix where rows represent observations and columns represent features. What does each row represent?

- A. A feature
- B. An observation/sample
- C. A matrix transformation
- D. An eigenvector

48. In a dataset matrix, each column generally represents:

- A. An observation
- B. A feature/variable
- C. An eigenvalue
- D. A class boundary

49. Suppose two features contain almost exactly the same information. What is the most likely consequence?

- A. Increased redundancy
- B. Increased independence
- C. Guaranteed increase in useful information
- D. Orthogonality

50. A dataset contains 100 features, but many of them are strongly correlated. What might this suggest mathematically?

- A. The data may have fewer effective independent directions than 100
- B. The data must have exactly 100 independent directions
- C. The data cannot be represented as a matrix
- D. All features are orthogonal

PART J — THINK BEFORE YOU ANSWER

51. A transformation changes a vector from $[2, 1]$ to $[6, 3]$. What can you immediately conclude?

- A. The direction has changed
- B. The vector has been scaled by 3
- C. The vector has become orthogonal
- D. The vector has been translated

52. Suppose three vectors lie along the same line. Their maximum number of linearly independent directions is:

- A. 0
- B. 1
- C. 2
- D. 3

53. If a matrix has highly dependent columns, its rank will generally be:

- A. Higher
- B. Lower
- C. Infinite
- D. Unaffected in all cases

54. A vector has a large magnitude but points in the same direction as another vector. What can be said about the two vectors?

- A. They must be orthogonal
- B. They may be scalar multiples of each other
- C. They must be independent
- D. They must have zero dot product

55. Which mathematical concept is most directly related to finding important directions in a linear transformation?

- A. Eigenvectors
- B. Mean
- C. Median
- D. Mode